

1. From sales to profit

MANAGMNT 494BI | September 16, 2026 | Work individually | 35–45 minutes total

Before choosing a strategy, learn to explain the numbers. This exercise uses invented results for one simplified sensor company. It is not a real Courier or a set of live Capstone decisions. A calculator is sufficient; a spreadsheet is optional. Keep your calculations and explain what each result means.

Use the [Capstone 2.0 Glossary & Guidance at langenkamp.io/capstone/](https://langenkamp.io/capstone/). Dollar amounts below are in \$ thousands unless marked per unit. Quantities are in thousands of units. Thus 800 means 800,000 sensors; a \$30 price remains \$30 per sensor.

Income statement input	Amount
Units sold / selling price	800 thousand / \$30 per unit
Material cost of units sold	8,000
Labor cost of units sold	5,000
Inventory carrying expense	480
Annual depreciation	1,600
SG&A: R&D + promotion + sales + administration	720 + 1,200 + 1,200 + 800
Other operating expenses / profit sharing	0 / 0
Interest expense / tax expense	1,000 / 1,200

A. Build the income statement (12 minutes)

Calculate, in order: (1) sales revenue; (2) total variable costs including carrying expense; (3) contribution margin dollars; (4) contribution margin percentage; (5) total SG&A; (6) EBIT; (7) net profit; and (8) return on sales. Write the formula beside each answer. For this exercise use the supplied tax expense; do not invent a tax rate.

Calculate material-plus-labor cost per unit and unit contribution before carrying costs. Why is this unit contribution different from contribution per unit after carrying costs? Why is neither figure net profit per unit?

B. Explain depreciation (5 minutes)

For a separate simple asset example, the company purchases \$24,000 thousand of equipment, with zero residual value and a 15-year straight-line life. Assume a full year of depreciation. Calculate annual depreciation. If paid entirely in cash at purchase, how much cash leaves at purchase? Is depreciation another payment each year?

Classify each item: revenue, operating expense, asset purchase, or financing transaction. (a) Selling sensors; (b) advertising; (c) buying an assembly line; (d) borrowing from a bank; (e) depreciation; (f) repaying loan principal. Explain why borrowing is not revenue and principal repayment is not interest expense.

2. Profit is not the cash balance

Use the net profit you calculated on page 1

C. Read the balance sheet (6 minutes)

Year-end balance sheet item	Amount (\$000)
Cash / accounts receivable / inventory	3,000 / 3,000 / 4,000
Gross plant and equipment	24,000
Accumulated depreciation (deduction)	6,000
Accounts payable / current debt	3,000 / 4,000
Long-term debt	7,000
Common stock / retained earnings	10,000 / 4,000

Calculate current assets, net plant, total assets, total liabilities and total equity. Show that assets = liabilities + equity. Treat accumulated depreciation as a deduction, not a second liability. Does \$4,000 thousand retained earnings mean the company has that much cash set aside?

Calculate net working capital and the current ratio, using accounts payable plus current debt as current liabilities here. Explain why a current ratio above 1 does not guarantee that tomorrow's bills can be paid.

D. Reconcile cash (8 minutes)

Cash-flow information for the year	Amount (\$000)
Beginning cash	4,000
Net profit / depreciation	Your page 1 result / 1,600
Increase in accounts receivable	800
Increase in inventory	1,200
Increase in accounts payable	400
Plant purchases paid in cash this year	6,000
New borrowing / principal repaid	3,000 / 800
Dividends / new shares / other cash adjustments	0 / 0 / 0

Calculate cash from operations using net profit + depreciation – increase in receivables – increase in inventory + increase in payables. Then calculate investing cash flow, financing cash flow, net change in cash and ending cash. Reconcile your result to the balance sheet above.

Explain in one sentence each: Why is depreciation added back? Why does an increase in inventory reduce operating cash? Why can this profitable company finish with less cash than it started with? The full-year depreciation amount covers its asset portfolio; do not recalculate it solely from this year's \$6,000 thousand purchases. The separate asset example on page 1 is not an extra purchase to add here.

3. Read the ratios, then plan production

E. Calculate the Courier ratios (6 minutes)

Use your sales and net profit from page 1 and year-end assets/equity from page 2. Use these denominators consistently. Report percentages to two decimal places and turnover/leverage as times.

Measure	Formula to apply
Return on sales (ROS)	Net profit ÷ sales
Asset turnover	Sales ÷ total assets
Return on assets (ROA)	Net profit ÷ total assets
Leverage (equity multiplier)	Total assets ÷ total equity
Return on equity (ROE)	Net profit ÷ total equity
SG&A-to-sales	SG&A ÷ sales

Check that $\text{ROS} \times \text{asset turnover} = \text{ROA}$, and $\text{ROA} \times \text{leverage} = \text{ROE}$, using unrounded decimals. Which ratio describes profit per dollar of sales? Which describes sales per dollar of assets? Which can rise when equity shrinks even if operations have not improved?

F. A separate next-round production decision (5 minutes)

For this planning problem only, assume opening inventory of 100 thousand units, forecast sales of 900 thousand, desired ending inventory of 100 thousand and first-shift capacity of 800 thousand. Calculate production required and utilization relative to first-shift capacity. Will second-shift production be needed? Do not infer unit inventory from the earlier company balance sheet; these are separate planning assumptions.

If actual sales are only 750 thousand units but you produce the planned amount, what is ending inventory? Explain the effect on cash needs and carrying costs. Can capacity bought during this decision round solve a current-round capacity shortfall?

G. Exit explanation (3 minutes)

In 60–100 words, explain the difference among revenue, contribution, net profit and cash. Include one way a company can increase sales yet worsen its finances. Use at least two numbers from your work.

Before you finish

Label units. Show formulas. Do not count SG&A component budgets twice. Do not call a loan revenue. Distinguish expenses from cash payments. A positive profit or an impressive ratio does not replace a cash check.

Reference: langenkamp.io/capstone/ and the linked official Capstone 2.0 Team Member Guide. The glossary is a teaching companion; your configured simulation governs live decisions. Your instructor will specify collection method and deadline. No team work is required.